



MODUL PRAKTIKUM

ADMINISTRASI BASIS DATA

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MODUL PRAKTIKUM 4

Autentikasi & Server Roles pada SQL Server

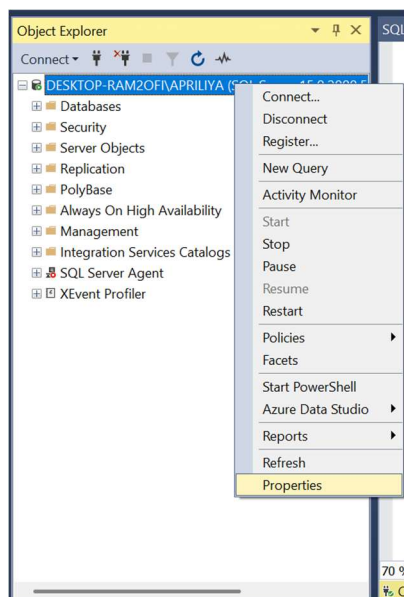
Tujuan Praktikum Agar Mahasiswa mampu:

1. Mengecek mode autentikasi SQL Server.
2. Mengaktifkan Mixed Mode Authentication.
3. Membuat login SQL Server.
4. Membuat login Windows.
5. Mengubah password login.
6. Menonaktifkan dan mengaktifkan login.
7. Memberikan Fixed Server Roles.
8. Menghapus role dari login.
9. Menampilkan daftar login dan role.

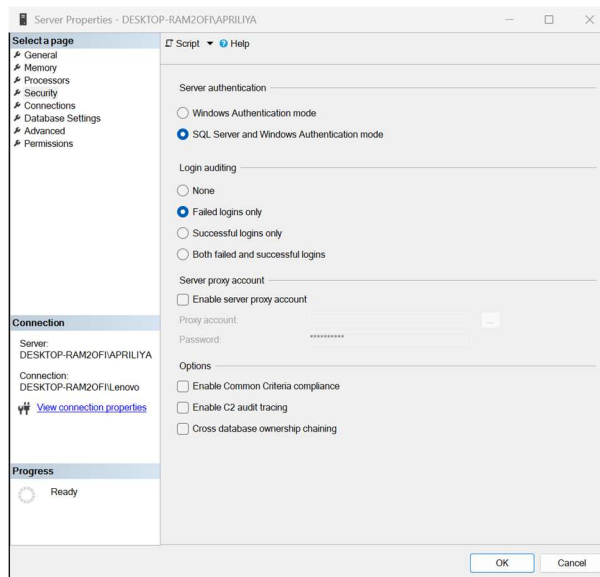
Praktikum 1 – Mengecek Authentication Mode

Langkah Kerja

1. Buka SQL Server Management Studio.
2. Login menggunakan Windows Authentication.
3. Klik kanan nama server.
4. Pilih Properties.



5. Pilih menu Security.



6. Perhatikan bagian Server Authentication.

Hasil yang Diharapkan

Mengetahui apakah server menggunakan:

- Windows Authentication Mode
- SQL Server and Windows Authentication Mode

Praktikum 2 – Mengaktifkan Mixed Mode Authentication

Langkah Kerja

1. Klik kanan nama server.
2. Pilih Properties.
3. Pilih tab Security.
4. Pilih SQL Server and Windows Authentication mode.
5. Klik OK.
6. Restart SQL Server.

Dokumentasi!!

Ambil screenshot:

- Sebelum perubahan
- Sesudah perubahan

Praktikum 3 – Membuat Login SQL Server

Query

```
SQLQuery1.sql - D...M2OFI\Lenovo (52))*
CREATE LOGIN operator_lab
WITH PASSWORD = 'Operator123!';
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:25:09.3997894+07:00

Verifikasi

```
SELECT name, type_desc
FROM sys.server_principals
WHERE name = 'operator_lab';
```

70 %

Results Messages

	name	type_desc
1	operator_lab	SQL_LOGIN

Hasil yang Diharapkan

Login operator_lab berhasil dibuat.

Praktikum 4 – Membuat Login Windows

Query

```
CREATE LOGIN [DESKTOP-RAM2OFI\aprilia]
FROM WINDOWS;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:32:57.9522269+07:00

Catatan

- Ganti NAMA-KOMPUTER dengan nama komputer masing-masing.
- Ganti User dengan akun Windows yang tersedia.

Verifikasi

```
SELECT name, type_desc
FROM sys.server_principals
WHERE name LIKE '%aprilia%';
```

70 %

Results Messages

	name	type_desc
1	Aprilia	SQL_LOGIN
2	DESKTOP-RAM2OFI\aprilia	WINDOWS_LOGIN

Praktikum 5 – Mengubah Password Login

Query

```
ALTER LOGIN operator_lab  
WITH PASSWORD = 'PasswordBaru123!';
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:37:34.0510396+07:00

Hasil yang Diharapkan

Password login operator_lab berhasil diubah.

Praktikum 6 – Menonaktifkan dan Mengaktifkan Login

Menonaktifkan Login

```
ALTER LOGIN operator_lab DISABLE;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:38:45.4278652+07:00

Mengaktifkan Login

```
ALTER LOGIN operator_lab ENABLE;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:39:51.0129857+07:00

Verifikasi

```
SELECT name, is_disabled  
FROM sys.sql_logins  
WHERE name = 'operator_lab';
```

70 %

Results Messages

	name	is_disabled
1	operator_lab	0

Keterangan

- 1 = Login nonaktif
- 0 = Login aktif

Praktikum 7 – Memberikan Server Role

Memberikan Role dbcreator

```
ALTER SERVER ROLE dbcreator
ADD MEMBER operator_lab;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:42:17.5833981+07:00

Memberikan Role securityadmin

```
ALTER SERVER ROLE securityadmin
ADD MEMBER operator_lab;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:43:16.5090909+07:00

Verifikasi

```
SELECT
    sp.name AS LoginName,
    sr.name AS ServerRole
FROM sys.server_role_members srm
JOIN sys.server_principals sp
    ON srm.member_principal_id = sp.principal_id
JOIN sys.server_principals sr
    ON srm.role_principal_id = sr.principal_id
WHERE sp.name = 'operator_lab';
```

70 %

Results Messages

	LoginName	ServerRole
1	operator_lab	securityadmin
2	operator_lab	dbcreator

Hasil yang Diharapkan

operator_lab memiliki role dbcreator dan securityadmin.

Praktikum 8 – Menghapus Server Role

Query

```
ALTER SERVER ROLE securityadmin
DROP MEMBER operator_lab;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-01T07:45:26.1444244+07:00

Verifikasi

```
SELECT
    sp.name AS LoginName,
    sr.name AS ServerRole
FROM sys.server_role_members srm
JOIN sys.server_principals sp
    ON srm.member_principal_id = sp.principal_id
JOIN sys.server_principals sr
    ON srm.role_principal_id = sr.principal_id
WHERE sp.name = 'operator_lab';
```

70 %

Results Messages

	LoginName	ServerRole
1	operator_lab	dbcreator

Hasil yang Diharapkan

Role securityadmin sudah tidak dimiliki operator_lab.

Praktikum 9 – Menampilkan Semua Login

Query

Berdasarkan nama

```
SELECT
    name,
    type_desc,
    create_date,
    modify_date
FROM sys.server_principals
ORDER BY name;
```

	name	type_desc	create_date	modify_date
1	##MS_AgentSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:58.890	2019-09-24 14:21:58.897
2	##MS_PolicyEventProcessingLogin##	SQL_LOGIN	2019-09-24 14:21:53.563	2022-10-04 14:39:15.223
3	##MS_PolicySigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.393	2019-09-24 14:21:16.393
4	##MS_PolicyTsqlExecutionLogin##	SQL_LOGIN	2019-09-24 14:21:53.570	2022-10-04 14:39:15.217
5	##MS_SmoExtendedSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.393	2019-09-24 14:21:16.393
6	##MS_SQLAuthenticatorCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.390	2019-09-24 14:21:16.390
7	##MS_SQLReplicationSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.390	2019-09-24 14:21:16.390
8	##MS_SQLResourceSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.390	2019-09-24 14:21:16.390
9	admin_lab	SQL_LOGIN	2026-04-01 07:54:48.943	2026-04-01 07:55:00.990
10	AdminUser	SQL_LOGIN	2025-03-03 04:22:03.680	2025-03-03 04:22:03.693
11	agent_A	SQL_LOGIN	2024-05-05 22:39:27.120	2024-12-09 22:04:13.097
12	Amni	SQL_LOGIN	2023-03-28 09:14:16.367	2023-03-28 09:14:16.380
13	Aprilia	SQL_LOGIN	2023-03-27 10:10:29.310	2023-03-27 10:10:29.323
14	bulkadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
15	dbcreator	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
16	Demologin	SQL_LOGIN	2023-03-27 11:56:41.993	2023-03-27 11:56:42.007
17	DESKTOP-RAM2OFI\april	WINDOWS_LOGIN	2026-04-01 07:32:57.933	2026-04-01 07:32:57.940
18	DESKTOP-RAM2OFI\Lenovo	WINDOWS_LOGIN	2022-10-04 14:39:14.563	2022-10-04 14:39:14.573
19	Dhewa	SQL_LOGIN	2023-03-28 15:49:04.463	2023-03-28 15:49:04.477
20	diskadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030

Query executed successfully. DESKTOP-RAM2OFI\APRILIYA (1... DESKTOP-RAM2OFI\Lenovo... master 00:00:00 43 rows

Berdasarkan tanggal pembuatan

```
SELECT
    name,
    type_desc,
    create_date,
    modify_date
FROM sys.server_principals
ORDER BY create_date;
```

	name	type_desc	create_date	modify_date
1	sa	SQL_LOGIN	2003-04-08 09:10:35.460	2025-06-14 22:58:11.290
2	public	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
3	sysadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
4	securityadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
5	serveradmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
6	setupadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
7	processadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
8	diskadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
9	dbcreator	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
10	bulkadmin	SERVER_ROLE	2009-04-13 12:59:06.030	2009-04-13 12:59:06.030
11	##MS_SQLResourceSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.390	2019-09-24 14:21:16.390
12	##MS_SQLReplicationSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.390	2019-09-24 14:21:16.390
13	##MS_SQLAuthenticatorCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.390	2019-09-24 14:21:16.390
14	##MS_PolicySigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.393	2019-09-24 14:21:16.393
15	##MS_SmoExtendedSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:16.393	2019-09-24 14:21:16.393
16	##MS_PolicyEventProcessingLogin##	SQL_LOGIN	2019-09-24 14:21:53.563	2022-10-04 14:39:15.223
17	##MS_PolicyTsqlExecutionLogin##	SQL_LOGIN	2019-09-24 14:21:53.570	2022-10-04 14:39:15.217
18	##MS_AgentSigningCertificate##	CERTIFICATE_MAPPED_LOGIN	2019-09-24 14:21:58.890	2019-09-24 14:21:58.897
19	DESKTOP-RAM2OFI\Lenovo	WINDOWS_LOGIN	2022-10-04 14:39:14.563	2022-10-04 14:39:14.573
20	NT.SERVICE\SQL Writer	WINDOWS_LOGIN	2022-10-04 14:39:14.580	2022-10-04 14:39:14.590

Query executed successfully. DESKTOP-RAM2OFI\APRILIYA (1... DESKTOP-RAM2OFI\Lenovo... master 00:00:00 43 rows

Hasil yang Diharapkan

Menampilkan seluruh login yang ada pada server.

Praktikum 10 – Menampilkan Semua Anggota Server Role

Query

```
SELECT
    sp.name AS LoginName,
    sr.name AS ServerRole
FROM sys.server_role_members srm
JOIN sys.server_principals sp
    ON srm.member_principal_id = sp.principal_id
JOIN sys.server_principals sr
    ON srm.role_principal_id = sr.principal_id
ORDER BY sr.name, sp.name;
```

70 %

Results Messages

	LoginName	ServerRole
1	operator_lab	dbcreator
2	DESKTOP-RAM2OFI\Lenovo	sysadmin
3	NT Service\MSSQL\$APRILIYA	sysadmin
4	NT SERVICE\SQLAgent\$APRILIYA	sysadmin
5	NT SERVICE\SQLWriter	sysadmin
6	NT SERVICE\Winmgmt	sysadmin
7	sa	sysadmin

Studi Kasus

Sebuah laboratorium memiliki kebutuhan berikut:

- admin_lab memiliki hak penuh.
- operator_lab hanya boleh membuat database.
- teknisi_lab hanya boleh mengelola login.

Query

```
CREATE LOGIN admin_lab
WITH PASSWORD = 'AdminLab123!';

CREATE LOGIN operator_lab2
WITH PASSWORD = 'OperatorLab123!';

CREATE LOGIN teknisi_lab
WITH PASSWORD = 'TeknisiLab123!';

ALTER SERVER ROLE sysadmin
ADD MEMBER admin_lab;

ALTER SERVER ROLE dbcreator
ADD MEMBER operator_lab2;

ALTER SERVER ROLE securityadmin
ADD MEMBER teknisi_lab;
```

Tugas Praktikum

1. Buat login baru bernama dosen_lab.
2. Berikan role dbcreator kepada dosen_lab.
3. Ubah password dosen_lab.
4. Tampilkan seluruh role yang dimiliki dosen_lab.
5. Nonaktifkan login dosen_lab.
6. Aktifkan kembali login dosen_lab.
7. Hapus role dbcreator dari dosen_lab.
8. Hapus login dosen_lab.

Format Pengumpulan

Mahasiswa wajib mengumpulkan:

1. File query SQL.
2. Screenshot hasil query.
3. Screenshot pengaturan authentication mode.
4. Analisis singkat hasil praktikum.
5. Kesimpulan.